

Institute of Engineering and Technology,
JK Lakshmipat University, Jaipur

Organizes

"JKLU AI-Buildathon 1.0"

("Build Autonomous Solutions for Everyday Campus Challenges.")

(16th November 2026)

Welcome to AI Buildathon 2026

Building Autonomous AI Solutions for Real-World Campus Challenges

Artificial Intelligence is shifting rapidly from static chat interfaces to autonomous AI agents—intelligent systems capable of reasoning, planning multi-step workflows, using complex tools, and taking meaningful actions.

AI Buildathon is our university's premier innovation challenge designed to bring this cutting-edge paradigm to our own campus. We are calling upon the brightest student innovators, developers, and thinkers to move beyond simple prompts and build next-generation AI agents that tackle real-world administrative, academic, operational, and student-life challenges right here at our university.

Why Participate?

- **Solve Real Campus Problems:** Work on actual operational bottlenecks—from exam scheduling, cybersecurity triage, and student support to financial reconciliation, HR onboarding, and admissions nurturing.
- **Master the Agentic Tech Stack:** Gain hands-on experience combining Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), tool calling, API integrations, and workflow automation.
- **Impact University Life:** The top solutions won't just sit in a repository they have the potential to be integrated into institutional workflows, transforming how our university operates.
- **Best 3 AI Agents get chance to win "Exciting Cash Prizes/Goodies worth 5 K"**
- Whether you are passionate about full-stack development, machine learning, systems architecture, or user experience design, AI- Build Athon provides the ideal stage to turn your ideas into autonomous, working prototypes.

Build Autonomous Solutions for Everyday Campus Challenges.

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Problem Statements

PS-01 · Unified Campus Support and Grievance Resolution Agent
PS-02 · Smart Campus Venue, Guest House and Parking Management Agent
PS-03 · Intelligent HR and Institutional Onboarding Agent
PS-04 · Intelligent Admission Enquiry and Lead Nurturing Agent
PS-05 · Intelligent Exam Scheduling, Invigilation and Seating Allocation Agent
PS-06 · Intelligent Placement Readiness and Career Matching Agent
PS-07 · Intelligent Cybersecurity SOC Triage and Incident Response Agent
PS-08 · MindEase — Agentic AI Companion for Emotional Wellbeing
PS-09: Intelligent Accounts Payable & Vendor Invoice Automation Agent

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PS-01 · Unified Campus Support and Grievance Resolution Agent

Field	Detail
Domain	Student Services, IT and Campus Administration.
Problem	Students repeatedly raise IT and administrative queries while hostel and campus grievances are scattered across calls, registers and messaging groups. These issues often lack centralised tracking, clear ownership, timely resolution and accountability.
Objective	Build an agent that serves as a unified campus support system, resolves routine queries using verified knowledge sources, records grievances, routes them to the appropriate department, tracks service-level agreements (SLAs), and escalates unresolved issues.
Suggested stack	LLM, RAG over campus SOPs and FAQs, web chat or WhatsApp intake, ticketing and workflow engine (n8n or custom), image understanding, PostgreSQL, SLA tracker
Deliverables	Unified conversational agent, automated query resolution and grievance intake module, ticket routing and SLA engine, student status dashboard, administrator dashboard, monthly issue-trend report
Expected outcome	Faster resolution of routine queries, transparent grievance tracking, reduced administrative workload, improved accountability and insights into recurring campus issues.

PS-02 · Smart Campus Venue, Guest House and Parking Management Agent

Field	Detail
Domain	Campus Administration and Facility Management.
Problem	Booking classrooms for activities outside their pre-slotted academic timetable, reserving amphitheatres and laboratories, managing MDC guest house availability, and coordinating visitor parking are handled through disconnected processes. Organisers often struggle to identify genuinely free rooms because they must cross-check the timetables of multiple semesters, while avoiding clashes with scheduled classes, lab sessions, events and existing reservations.
Objective	Build an agentic AI system that analyses the current semester timetables across all semesters to identify classrooms that are free during requested time slots, including classes not occupied by their pre-slotted lectures. The agent must also manage bookings for both amphitheatres, eligible laboratories, MDC guest house rooms and campus parking. It should check availability, detect conflicts, recommend alternatives and coordinate related bookings through a unified interface.
Suggested stack	LLM, React, FastAPI, PostgreSQL, timetable parser (PDF/Excel/CSV), scheduling and constraint-checking engine, calendar and booking APIs, workflow engine (n8n or custom), optional RAG for campus booking policies
Deliverables	Timetable-aware venue booking agent, free-classroom discovery module, amphitheatre and lab reservation module, MDC guest room availability and booking module, parking availability and allocation module, conflict detection and alternative recommendation engine, administrative dashboard and venue utilisation reports
Expected outcome	Accurate identification of free classrooms across all semesters, fewer scheduling conflicts, easier booking of classrooms for extra lectures, quizzes, workshops and events, better utilisation of amphitheatres and laboratories, and coordinated guest accommodation and visitor parking.

PS-03 · Intelligent HR and Institutional Onboarding Agent

Field	Detail
Department	IT and HR Analytics.
Problem	Onboarding new students, faculty and staff involves multiple manual steps for account creation, email, ERP/SIS access and Wi-Fi provisioning. Meanwhile, HR data on recruitment, attrition, absenteeism and training is often analysed manually, delaying management insights and decisions.
Objective	Build an agentic AI platform that automates onboarding workflows using role-based access rules and approval gates, while analysing approved HR datasets to identify trends, validate figures and generate evidence-based management summaries.
Suggested stack	LLM, React, FastAPI, PostgreSQL, Microsoft Graph API (sandbox/developer tenant), mock ERP/SIS and Wi-Fi APIs, Python/pandas, SQL tools, charting library, workflow engine
Deliverables	Onboarding request portal, role-based provisioning agent, IT approval dashboard, onboarding status tracker, HR analytics agent, trend dashboard, validated management report
Expected outcome	Faster and more consistent onboarding, fewer provisioning errors, controlled access assignment, quicker HR reporting and more reliable evidence-based administrative decisions.

PS-04 · Intelligent Admission Enquiry and Lead Nurturing Agent

Field	Detail
Domain	Admissions and Student Recruitment
Problem	Universities receive a high volume of admission enquiries during peak seasons across websites, email, and messaging platforms. Manual responses, inconsistent information, missed follow-ups, and delayed counsellor engagement can result in losing prospective students. Existing systems often record leads but do not actively manage the entire enquiry-to-counselling journey.
Objective	Build an agentic AI system that answers admission enquiries using verified institutional information, qualifies prospective students, maintains CRM records, schedules counsellor appointments, and adapts follow-up actions based on each lead's responses and engagement.
Suggested stack	LLM, RAG, React, FastAPI, PostgreSQL, mock CRM or Salesforce Developer sandbox, simulated WhatsApp/email APIs, calendar API, workflow orchestration.
Deliverables	Multi-channel enquiry interface, knowledge-grounded response agent, lead qualification and scoring engine, CRM integration, appointment scheduling, personalised follow-up workflow, counsellor dashboard, and action audit log.
Expected outcome	Faster and more consistent enquiry handling, better lead prioritisation, fewer missed follow-ups, efficient counsellor scheduling, and a traceable admission enquiry workflow.

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PS-05 · Intelligent Exam Scheduling, Invigilation and Seating Allocation Agent

Field	Detail
Domain	Examinations and Academic Operations
Problem	Manually preparing examination timetables, assigning invigilators, allocating students to venues, and generating seating plans is time-consuming and prone to errors. In addition to scheduling clashes and room capacity limits, seating arrangements must minimise the proximity of students appearing for the same course to reduce opportunities for collusion. Last-minute changes make the process even more complex.
Objective	Build an agentic AI system that generates clash-free examination timetables, assigns invigilators, allocates students to suitable venues, and produces seat-level arrangements that maximise separation between students taking the same course. The agent must support natural-language changes, validate constraints, explain conflicts, and publish plans only after administrator approval.
Suggested stack	LLM, Google OR-Tools constraint solver, Python, FastAPI, React, PostgreSQL or CSV-based mock data, PDF/Excel export libraries.
Deliverables	Timetable generation agent, invigilation allocation engine, venue allocation module, seat-level seating planner, conflict detection and explanation module, natural-language editing interface, approval dashboard, and PDF/Excel exports.
Expected outcome	A validated examination timetable, balanced invigilation assignments, efficient venue utilisation, and seating plans that maximise separation between students taking the same course.

PS-06 · Intelligent Placement Readiness and Career Matching Agent

Field	Detail
Domain	Careers, Training and Placement
Problem	Students often apply for jobs without understanding how well their skills align with role requirements or what they need to improve before placement season. Meanwhile, placement cells lack a consolidated, evidence-based view of student readiness, recurring skill gaps, and preparation progress across different job roles.
Objective	Build an agentic AI system that matches students with suitable job opportunities, identifies and prioritises skill gaps, creates personalised upskilling plans, and conducts adaptive mock interviews. The agent should track progress over time and generate aggregated insights to help the placement cell improve training programmes.
Suggested stack	LLM, resume parsing, embeddings and semantic search, React, FastAPI, PostgreSQL, mock job-description dataset, speech-to-text and text-to-speech APIs, Python-based skill-matching engine, course/resource catalogue.
Deliverables	Student career dashboard, job-matching agent, explainable skill-gap analyser, personalised upskilling planner, adaptive mock interview module, progress tracker, and placement-cell analytics dashboard.
Expected outcome	Better-informed job applications, targeted preparation, measurable progress in placement readiness, and actionable insights into cohort-wide skill gaps.

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PS-07 · Intelligent Cybersecurity SOC Triage and Incident Response Agent

Field	Detail
Domain	Cybersecurity and Security Operations
Problem	Security Operations Centre (SOC) teams receive large volumes of firewall, authentication, and endpoint alerts, making it difficult to distinguish genuine threats from false positives. Manual investigation delays incident response, while insufficient context can lead to incorrect severity assessments and unnecessary containment actions.
Objective	Build an agentic AI system that investigates security alerts, enriches threat indicators, correlates related events, prioritises incidents, and recommends evidence-based response actions for analyst approval. The agent should adapt its investigation when evidence is incomplete, contradictory, or indicates a higher-risk incident.
Suggested stack	LLM, Python, FastAPI, React, Wazuh or ELK with sample logs, MITRE ATT&CK framework, VirusTotal/AbuseIPDB APIs or cached threat-intelligence data, PostgreSQL.
Deliverables	Alert investigation agent, event correlation engine, severity scoring system, threat-intelligence enrichment module, incident dashboard, analyst approval workflow, response simulation engine, and automatically drafted incident reports.
Expected outcome	Reduced alert fatigue, faster investigations, better incident prioritisation, traceable response recommendations, and consistent human-approved incident handling.

PS-08 · MindEase — Agentic AI Companion for Emotional Wellbeing

Field	Detail
Domain	Psychology, Student Wellbeing and Emotional Support
Problem	Students frequently experience academic pressure, demotivation, loneliness, emotional exhaustion, and sadness. They may want someone to talk to without being judged, receiving unsolicited advice, or being told to simply work harder. Conventional chatbots often respond with repetitive motivational statements and fail to maintain meaningful conversations or adapt to the user's emotional needs.
Objective	Build an agentic AI companion that engages users in natural, empathetic, ongoing conversations about their feelings, daily experiences, academic stress, relationships, and personal challenges. The agent should listen, ask thoughtful follow-up questions, help users explore their feelings, and offer practical support when wanted. A smaller set of optional psychology-informed activities, interactive stories, creative exercises, and manageable micro-tasks should complement the conversation rather than dominate it.
Suggested stack	LLM, React, FastAPI, PostgreSQL or SQLite, RAG over curated wellbeing resources, consent-based conversational memory, streaming chat interface, optional speech-to-text and text-to-speech, and an agent workflow engine.
Deliverables	Conversational AI companion, context-aware dialogue engine, personalised follow-up system, optional voice conversation, user-controlled memory, on-demand activity recommendations, wellbeing resource navigator, and safety-aware human support referral workflow.
Expected outcome	An approachable, conversational companion that helps users feel heard, reflect on their experiences, explore constructive ways forward, and access appropriate support when needed—without making every interaction about productivity or completing tasks.

PS-09. Intelligent Accounts Payable & Vendor Invoice Automation Agent

Field	Detail
Domain	Accounts Payable, Procurement, and Vendor Finance.
Problem	Universities work with dozens of external vendors, contractors, and service providers. Processing physical or PDF vendor invoices, cross-checking them against purchase orders (POs) and delivery logs, checking department budget limits, and securing multi-level administrative approvals is entirely manual, prone to human error, and causes delayed payments.
Objective	Build an agentic AI platform that ingests vendor invoices via email or a portal, uses document understanding to parse line items, performs automated three-way matching (Invoice vs. Purchase Order vs. Goods Receipt), flags discrepancies or duplicate billings, checks departmental budget availability, and routes verified invoices through approval chains.
Suggested stack	LLM, Vision API / Document OCR, React, FastAPI, PostgreSQL, Python for matching logic, mock payment and ERP ledger APIs, workflow engine.
Deliverables	Vendor portal intake module, OCR invoice parser, automated three-way matching engine, budget-limit validation checker, finance approval dashboard, and payment schedule tracker.
Expected outcome	Faster invoice processing, elimination of duplicate or over-billed payments, automated tracking of liabilities, and minimized manual intervention in accounts payable.

General Rules & Event Guidelines

1. Core Architectural & Agentic Principles

- **True Agentic Functionality Required:** Submissions must demonstrate true autonomous behavior—planning multi-step workflows, selecting and using tools (APIs, databases, web search, constraint solvers, OCR), verifying execution outputs, and recovering from errors. Simple LLM wrappers, static chatbots, or basic UI forms with no workflow automation will not qualify for top scoring.
- **Knowledge Grounding & Zero Hallucination:** Agents answering institutional questions (Admissions, SOPs, Financial Rules, Grant Guidelines) must retrieve information exclusively from provided datasets or approved knowledge sources. Inventing policies, fees, dates, or eligibility criteria is strictly prohibited.
- **Human-in-the-Loop & Approval Gates:** Any action involving state changes (e.g., publishing exam timetables, issuing account credentials, granting fee waivers, approving vendor payouts, or initiating security containment) must implement explicit human approval workflows before execution.

2. Development & Implementation Constraints

- **Sandbox & Synthetic Data Usage:** All integrations (CRMs, messaging channels, provisioning endpoints, payment portals) must use provided sandboxes or mock APIs. Real institutional systems, live student records, or production APIs must **not** be connected or modified under any circumstances.
- **Data Privacy & Security:**
    - No personally identifiable information (PII) beyond provided synthetic datasets should be collected or processed.
    - Role-Based Access Control (RBAC) must be enforced across administrator dashboards, student portals, and staff interfaces.
- **Traceability & Logging:** Systems must maintain audit logs recording agent reasoning steps, API calls attempted, tool outputs, and human approvals/rejections.

3. Deliverables

Students must submit their complete GitHub repository link, including all source code, project files, documentation, and relevant resources, before the evaluation scheduled for 15 November at 12:00 PM.

4. Team Size

Team size can be from 1 to 4. Interdepartmental teams are also allowed.

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Evaluation Criteria

1. Agentic Capabilities & Workflow Automation (30 Points)

- **Multi-Step Reasoning & Planning (10 pts):** Ability to decompose complex user prompts into logical execution plans.
- **Tool & API Integration (10 pts):** Effective and reliable use of tools (calculators, OCR engines, constraint solvers, database queries, mock APIs).
- **Adaptability & Error Recovery (10 pts):** Handling unexpected inputs, missing data, or tool failures gracefully (e.g., retrying failed API calls, asking clarifying questions, or recommending alternative options).

2. Domain Accuracy, Constraints & Problem Solving (25 Points)

- **Functional Correctness (10 pts):** Satisfying all domain-specific requirements (e.g., clash-free timetables in PS-05, accurate fee ledgers in PS-09, correct SLA routing in PS-01).
- **Knowledge Grounding (10 pts):** Strict adherence to provided SOPs, fee structures, and guidelines with zero fabricated rules or facts.
- **Optimization & Reasoning Quality (5 pts):** Quality of outputs (e.g., optimal venue search, seat separation, explainable skill-gap scoring, or incident timelines).

3. System Architecture & Technical Implementation (20 Points)

- **Stack Integration & Code Quality (10 pts):** Clean, modular backend and frontend development using the recommended or equivalent tech stacks.
- **Performance & Efficiency (5 pts):** Low latency in agent decision-making, streaming response handling, and computational efficiency.
- **Scalability & Data Handling (5 pts):** Proper database design, state management, and handling of multi-user operations.

4. Safety, Privacy & Human-in-the-Loop Governance (15 Points)

- **Approval Gates & Human Oversight (5 pts):** Enforcing administrator verification for sensitive actions before modifying database states.
- **Access Control & Privacy (5 pts):** Role-based access enforcement and secure handling of synthetic data.
- **Auditability & Explainability (5 pts):** Maintaining transparent action logs and clear reasoning traces for agent decisions.

5. Demonstration, UX & Presentation Quality (10 Points)

- **User Experience & Dashboard Design (5 pts):** Intuitive, clean interface for both end-users (students/staff) and administrators.
- **Live Demo & Clarity of Pitch (5 pts):** Effective demonstration of working end-to-end workflows during live evaluation.

You are free to use any tech stack / process flow as you wish. Let's get building.

Organizing Committee:

Dr. Akshat Agrawal
Associate Professor
IET, JKLU
(Akshat.agrawal@jklu.edu.in)

Student Organizer

Chandra Mauli Singh
(chandramaulisingh@jklu.edu.in)

+917665139766

Tejender Singh Shekhawat
(tejendersinghshekhawat@jklu.edu.in)

Tejendra Singh
(tejendrasingh@jklu.edu.in)